MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECSC303 - WEB DESIGNING USING PHP

(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 70

Scheme for Evaluation

Part A: Very Short Answer Questions (Answer all) - (5 * 2 = 10 Marks)

1. **Explain the purpose of the declaration** [U] [CO1]

   Answer: (1 mark each)

   - The <!DOCTYPE> declaration tells the browser which version/type of HTML the document is written in, so that the page is interpreted and rendered correctly (in standards mode) (1 mark).
   - It must appear as the very first line of the document, before the <html> tag; it is an instruction to the browser, not an HTML tag, and has no closing tag. Example: <!DOCTYPE html> (1 mark).

2. **Identify the syntax used to set text color using the <font> tag in HTML** [U] [CO1]

   Answer: (1 mark each)

   - Syntax: `<font color="value"> text </font>`, where value is a colour name or a hexadecimal RGB value #RRGGBB (1 mark).
   - Example: `<font color="red">Welcome</font>` or `<font color="#FF0000">Welcome</font>` (1 mark).

3. **Explain any two comparison operators in PHP with examples.** [U] [CO2]

   Answer: (1 mark each for any two operators with example)

   - == (equal): true if both values are equal after type conversion, e.g. `10 == "10"` is true (1 mark).
   - === (identical): true if the values are equal and of the same type, e.g. `10 === "10"` is false (1 mark).
   - Also accept != or <> (not equal), !== (not identical), <, >, <=, >= with a correct example.

4. **Describe the use of the rand() function in loops.** [U] [CO3]

   Answer: rand(min, max) returns a random integer between min and max (both inclusive); without arguments it returns a random integer between 0 and getrandmax() (1 mark). When called inside a loop, a fresh random number is produced in every iteration, e.g. to simulate dice throws or generate test data: `for ($i = 1; $i <= 5; $i++) { echo rand(1, 6) . " "; }` (1 mark).

5. **State the purpose of mysqli_query() in PHP.** [K] [CO4]

   Answer: mysqli_query() sends/executes an SQL query on the MySQL database through an open connection: `mysqli_query($conn, $sql)` (1 mark).

   For SELECT it returns a result set (mysqli_result object) that is read with mysqli_fetch_row/assoc/array; for INSERT, UPDATE, DELETE, CREATE it returns true on success; it returns false on failure (1 mark).

Part B: Short Answer Questions (Answer any 6 out of 8 Questions) - (6 * 5 = 30 Marks)

6. **Describe the steps involved in HTTP communication.** [U] [CO1]

   Answer: (1 mark for each step; any five)

   - The user enters a URL in the browser (client) or clicks a hyperlink (1 mark).
   - The domain name in the URL is resolved to the server's IP address using DNS (1 mark).
   - The browser opens a TCP/IP connection with the web server (default port 80) (1 mark).
   - The browser sends an HTTP request containing the method (GET/POST), the requested resource and request headers (1 mark).
   - The web server processes the request, locating the file or running a server-side script such as PHP (1 mark).
   - The server returns an HTTP response with a status code (e.g. 200 OK, 404 Not Found), response headers and the content (HTML, images, etc.) (1 mark).
   - The browser renders the page; the connection is closed since HTTP is a stateless request–response protocol (1 mark).

7. **Explain the structure of an HTML table using <table>, <tr>, <th>, and <td> tags, and provide a suitable example.** [U] [CO1]

   Answer: (1 mark for each point)

   - <table> … </table>: container that defines the table; attributes such as border, width, cellpadding, cellspacing, align (1 mark).
   - <tr> … </tr>: defines a table row; each row holds header or data cells (1 mark).
   - <th> … </th>: defines a header cell; its contents are bold and centred by default (1 mark).
   - <td> … </td>: defines a data cell; colspan/rowspan merge cells across columns/rows (1 mark).
   - Correct example table (1 mark).

Reference example:

```
<table border="1">
  <caption>Student Marks</caption>
  <tr>
    <th>Roll No</th> <th>Name</th> <th>Marks</th>
  </tr>
  <tr>
    <td>1</td> <td>Anu</td> <td>85</td>
  </tr>
  <tr>
    <td>2</td> <td>Rahul</td> <td>78</td>
  </tr>
</table>
```

8. **Demonstrate the use of background images and related CSS background properties with a code example.** [A] [CO2]

Answer: (1 mark for each point)

- background-image: url("file") sets an image as the background of an element (1 mark).
- background-repeat: repeat | repeat-x | repeat-y | no-repeat controls tiling of the image (1 mark).
- background-position (e.g. center, top left, x y) positions the image; background-attachment: scroll | fixed decides whether it scrolls with the page (1 mark).
- background-color as fallback; background-size or the background shorthand property (1 mark).
- Correct code example applying these properties (1 mark).

Reference program:

```
<html>
<head>
<style>
  body {
    background-color: #F0F0F0;
    background-image: url("bg.jpg");
    background-repeat: no-repeat;
    background-position: center top;
    background-attachment: fixed;
  }
</style>
</head>
<body>
  <h1>Welcome to Our College</h1>
</body>
</html>
```

9. **Explain the role of MySQL in XAMPP.** [U] [CO2]

Answer: (1 mark for each point)

- XAMPP is an integrated package: X (cross-platform), Apache (web server), MySQL/MariaDB (database server), PHP and Perl (1 mark).
- MySQL is the relational database management system in the package; it stores the data of web applications in databases and tables (1 mark).

- It is started/stopped from the XAMPP Control Panel and runs as a database server, by default on port 3306 (1 mark).
- phpMyAdmin, bundled with XAMPP, provides a browser-based interface to create databases and tables, run SQL and manage users (1 mark).
- PHP scripts running on Apache connect to MySQL (mysqli functions) to insert, retrieve, update and delete data, making the website dynamic and database-driven (1 mark).

10. Apply array_push() to add a new movie to a favorites list. [A] [CO3]

Answer: (Step marks: syntax 2, logic 2, output 1)

- Syntax: array_push($array, value1, value2, …) adds one or more elements to the end of the array and returns the new number of elements (1 mark).
- Correct PHP syntax of the program (array creation, statements, <?php … ?>) (1 mark).
- Logic: create the favorites list as an array (1 mark).
- Logic: call array_push() correctly to add the new movie (1 mark).
- Output: display the updated list using print_r(), var_dump() or a foreach loop, with correct output (1 mark).

Reference program:

```
<?php
$favorites = array("Inception", "Interstellar", "The Dark Knight");
$count = array_push($favorites, "Oppenheimer");
echo "Total movies: $count <br>";
foreach ($favorites as $movie) {
    echo $movie . "<br>";
}
?>
```

Output:

```
Total movies: 4
Inception
Interstellar
The Dark Knight
Oppenheimer
```

11. Illustrate the working of a multidimensional array in PHP with a suitable code example. [A] [CO3]

Answer: (1 mark for each point)

- A multidimensional array is an array whose elements are themselves arrays (an array of arrays); a two-dimensional array stores data in rows and columns (1 mark).
- Each element is accessed with two (or more) indices or keys, e.g. $students[1][2]; elements can be numeric, associative or mixed (1 mark).
- Syntax: correct declaration of the multidimensional array (1 mark).
- Logic: traversal using nested for/foreach loops (1 mark).
- Output: correct output of the program (1 mark).

Reference program:

```
<?php
$students = array(
    array("Anu", 85, 90),
    array("Rahul", 78, 88),
    array("Meera", 92, 95)
);
echo "<table border='1'>";
echo "<tr><th>Name</th><th>Mark 1</th><th>Mark 2</th></tr>";
for ($row = 0; $row < count($students); $row++) {
    echo "<tr>";
    for ($col = 0; $col < count($students[$row]); $col++) {
        echo "<td>" . $students[$row][$col] . "</td>";
    }
    echo "</tr>";
}
echo "</table>";
echo "Second mark of Rahul: " . $students[1][2];
?>
```

Output:

```
Name   Mark 1  Mark 2
Anu    85      90
Rahul  78      88
Meera  92      95
Second mark of Rahul: 88
```

12. Describe the steps involved in creating a table and then adding one record into it. [U] [CO4]

Answer: (1 mark for each step)

- Start the MySQL server (XAMPP Control Panel) and open the MySQL client, phpMyAdmin or a PHP connection using mysqli_connect() (1 mark).
- Create and select the database: `CREATE DATABASE college; USE college;` (1 mark).
- Create the table with CREATE TABLE, specifying column names, data types and constraints such as PRIMARY KEY / NOT NULL (1 mark).
- Add one record with INSERT INTO table (columns) VALUES (values); string values in quotes (1 mark).
- Execute and verify, e.g. run the statements through mysqli_query() and check its return value, and confirm the record using SELECT (1 mark).

Reference example:

```
CREATE DATABASE college;
USE college;
CREATE TABLE student (
    roll_no INT PRIMARY KEY,
    name    VARCHAR(50) NOT NULL,
    marks   INT
);
INSERT INTO student (roll_no, name, marks) VALUES (1, 'Anu', 85);
SELECT * FROM student;
```

13. **Examine why the database name is required in mysqli_connect() even though host and username are given.** [An] [CO4]

Answer: (1 mark for each point)

- Syntax: `mysqli_connect(host, username, password, database);` it returns a connection object, or false on failure (1 mark).
- Host and username (with password) only locate the MySQL server and authenticate the user; they do not identify which data is to be used (1 mark).
- One MySQL server holds many databases; the database name selects the default database on which all subsequent queries through that connection operate (1 mark).
- Without a selected database, queries on unqualified table names fail with a "No database selected" error; the database would then have to be chosen with mysqli_select_db($conn, "dbname") or every table written as dbname.table (1 mark).
- Giving it in mysqli_connect() connects and selects in one step, limits the script to the intended database and simplifies error checking with mysqli_connect_error(), typically in a reusable connection file (1 mark).

Reference example:

```
$conn = mysqli_connect("localhost", "root", "", "college");
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
$result = mysqli_query($conn, "SELECT * FROM student");
```

Part C: Essay Questions (Answer any 2 out of 3 Questions) - (2 * 15 = 30 Marks)

14. **Explain CSS and different types of CSS selectors.** [U] [CO2]

Answer: (CSS – 6 marks; Selectors – 9 marks)

CSS (6 marks)

- Definition: Cascading Style Sheets describe the presentation (fonts, colours, spacing, layout) of HTML documents, separating content from formatting (1 mark).
- Advantages: consistent look across pages, easy maintenance, reduced code, faster loading; cascading/inheritance of styles (1 mark).
- Rule syntax: `selector { property: value; }`, e.g. `h1 { color: navy; font-size: 24px; }` (1 mark).

- Inline style: style attribute on an element, e.g. `<p style="color:red">` (1 mark).
- Internal (embedded) style sheet: rules placed inside the <style> element in <head> (1 mark).
- External style sheet: rules in a .css file linked with `<link rel="stylesheet" href="style.css">` (1 mark).

CSS Selectors (9 marks)

- Concept: a selector identifies the HTML element(s) to which a style rule is applied (1 mark).

Any four of the following, 2 marks each (1 mark for explanation, 1 mark for example):

- Type (element) selector: selects all elements of a given tag – `p { color: blue; }`
- Class selector: selects elements with a class attribute; can be reused on many elements – `.highlight { background-color: yellow; }` used as `<p class="highlight">`
- ID selector: selects the single element with a given id – `#header { font-size: 24px; }` used as `<div id="header">`
- Universal selector: selects all elements – `* { margin: 0; }`
- Grouping selector: applies one rule to several selectors – `h1, h2, h3 { font-family: Arial; }`
- Descendant (contextual) selector: selects elements nested inside another – `div p { color: green; }`
- Pseudo-class selector: selects elements in a particular state, e.g. link pseudo-classes – `a:link, a:visited, a:hover, a:active, a:hover { color: red; }`
- Attribute selector / pseudo-element: `input[type="text"] { border: 1px solid gray; }, p::first-letter { font-size: 200%; }`

15. Explain different branching statements in PHP with examples. [U] [CO3]

Answer: (Introduction 1 mark; if 3; if…else 3; if…elseif…else 3; switch 4; ternary operator 1)

- Introduction: branching (decision-making) statements execute different blocks of code depending on whether a condition is true or false (1 mark).
- if statement – syntax 1, explanation 1, example 1 (3 marks).
- if…else statement – syntax 1, explanation 1, example 1 (3 marks).
- if…elseif…else statement (checks several conditions in sequence) – syntax 1, explanation 1, example 1 (3 marks).
- switch statement – syntax with case, break and default 1; explanation of matching an expression against case values 1; role of break (fall-through) and default 1; example 1 (4 marks).
- Ternary (conditional) operator `condition ? value1 : value2` as a compact if…else (1 mark).

Reference examples:

```
// if
$marks = 82;
if ($marks >= 40) {
    echo "Passed";
}

// if...else
if ($marks >= 40) {
    echo "Passed";
} else {
    echo "Failed";
}

// if...elseif...else
if ($marks >= 80) {
    echo "Grade A";
} elseif ($marks >= 60) {
    echo "Grade B";
} else {
    echo "Grade C";
}

// switch
$day = 3;
switch ($day) {
    case 1:  echo "Monday";    break;
    case 2:  echo "Tuesday";   break;
    case 3:  echo "Wednesday"; break;
    default: echo "Invalid day";
}

// ternary operator
$result = ($marks >= 40) ? "Pass" : "Fail";
echo $result;
```

16. **Illustrate the complete data retrieval and modification process for a Student table. (a) Explain SELECT and write queries to retrieve all records and only name and marks. (b) Write a SELECT query with WHERE to list students scoring more than 75. (c) Write an UPDATE and a DELETE query, explaining WHERE in both.** [A] [CO4]

Answer: (a) 6 marks; (b) 4 marks; (c) 5 marks. Assume Student(roll_no, name, marks).

(a) SELECT (6 marks)

- SELECT retrieves data (rows/columns) from one or more tables; the result is returned as a result set (1 mark).
- Syntax: `SELECT column1, column2 | * FROM table_name [WHERE condition] [ORDER BY column];` – * selects all columns (1 mark).
- Retrieve all records: `SELECT * FROM Student;` (2 marks).
- Retrieve only name and marks: `SELECT name, marks FROM Student;` (2 marks).

(b) SELECT with WHERE (4 marks)

- Query: `SELECT * FROM Student WHERE marks > 75;` (3 marks; also accept `SELECT name, marks FROM Student WHERE marks > 75;`).
- Explanation: WHERE filters the rows, so only students whose marks exceed 75 are listed (1 mark).

(c) UPDATE and DELETE (5 marks)

- UPDATE query: `UPDATE Student SET marks = 80 WHERE roll_no = 3;` – modifies existing values (2 marks).
- DELETE query: `DELETE FROM Student WHERE roll_no = 5;` – removes matching rows (2 marks).
- Role of WHERE in both: it restricts the change to the rows that satisfy the condition; if WHERE is omitted, UPDATE changes every row and DELETE removes all rows of the table (1 mark).

Optional PHP illustration of the process:

```
$conn = mysqli_connect("localhost", "root", "", "college");
$result = mysqli_query($conn, "SELECT name, marks FROM Student WHERE marks
> 75");
while ($row = mysqli_fetch_assoc($result)) {
    echo $row["name"] . " - " . $row["marks"] . "<br>";
}
mysqli_query($conn, "UPDATE Student SET marks = 80 WHERE roll_no = 3");
mysqli_query($conn, "DELETE FROM Student WHERE roll_no = 5");
mysqli_close($conn);
```